

Causal Inference for Health Data

(STATS C160/C260 — Winter 2026)

Lecture 9: Counterfactuals I

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Rules of Do-Calculus

Theorem. The following transformations are valid for any do-distribution induced by a SCM M :

Rule 1: Adding/removing Observations

$$P(y|do(x), \textcolor{brown}{z}, w) = P(y|do(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{G_{\bar{X}}}$$

Rule 2: Action/observation exchange

$$P(y|do(x), \textcolor{brown}{do(z)}, w) = P(y|do(x), \textcolor{brown}{z}, w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{G_{\bar{X}\underline{Z}}}$$

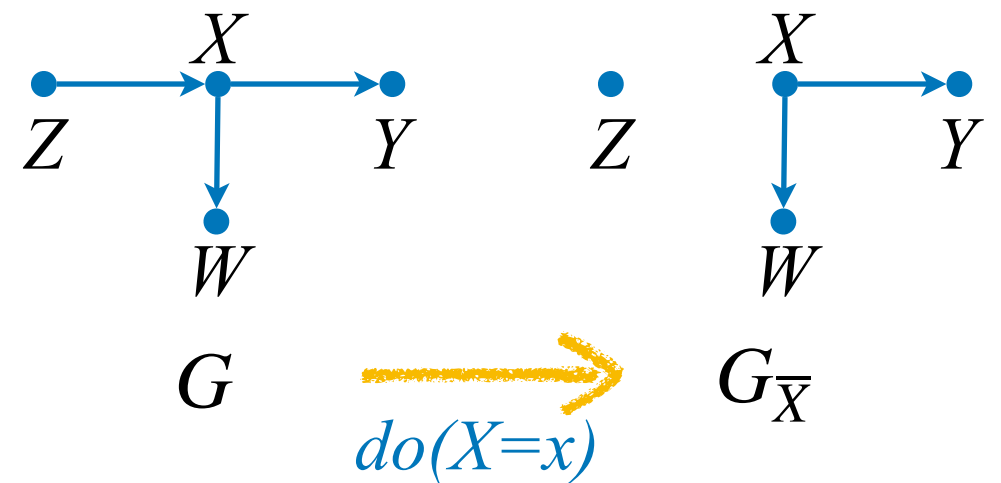
Rule 3: Adding/removing Actions

$$P(y|do(x), \textcolor{brown}{do(z)}, w) = P(y|do(x), w) \quad \text{if} \quad (Y \perp\!\!\!\perp Z \mid X, W)_{G_{\bar{X}\overline{Z(W)}}}$$

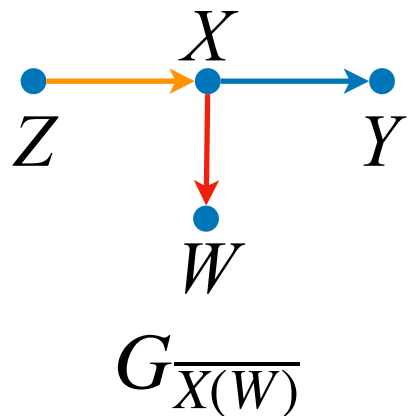
where $Z(W) = Z \setminus \text{an}(W)_{G_{\bar{X}}}$ is the subset of Z that are not ancestors of W in $G_{\bar{X}}$.

A curiosity about R3

- Regarding R3's contingency, note that the probability of Z given W may not be the same with or without intervening on X .



To witness, compare $G_{\bar{X}}$ and $G_{\overline{X(W)}}$:

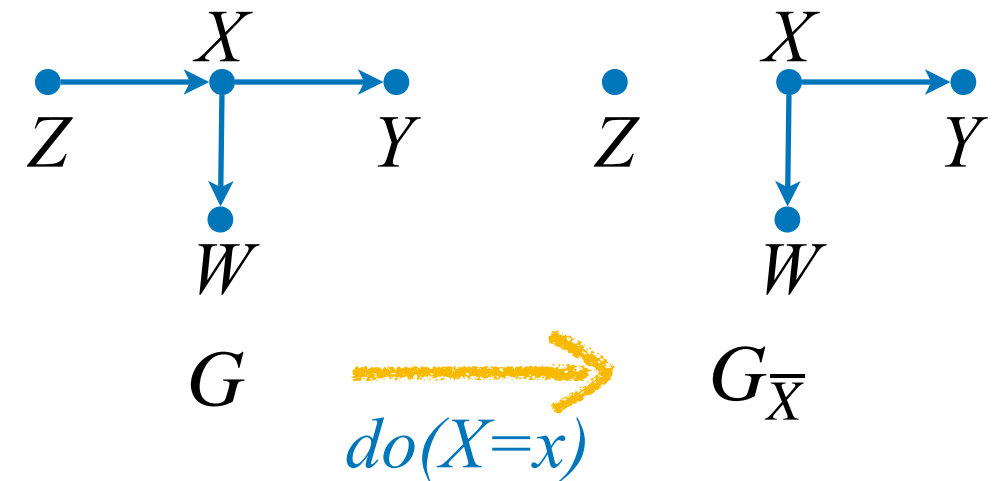


$$(Z \not\perp\!\!\!\perp X \mid W)_{G_{\overline{X(W)}}} \Rightarrow \exists_M P(z|do(x), w) \neq P(z|w)$$

What? 🙋

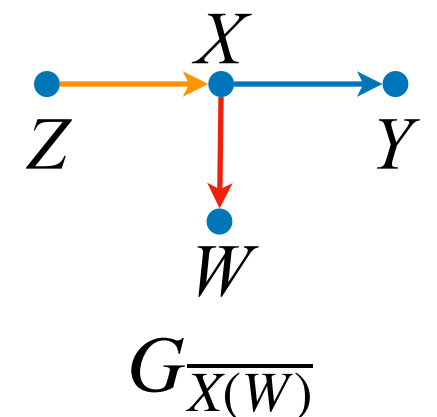
A curiosity about R3

- Verifying whether $P(z|do(x), w) \neq P(z|w)$ holds:



$$\begin{aligned}
 P(z|do(x), w) &= \frac{P(z, w | do(x))}{P(w | do(x))} \\
 &= \frac{\sum_y P(z)P(w | x)P(y | x)}{\sum_{y,z} P(z)P(w | x)P(y | x)} \\
 &= \frac{P(z)P(w | x)}{P(w | x)}
 \end{aligned}$$

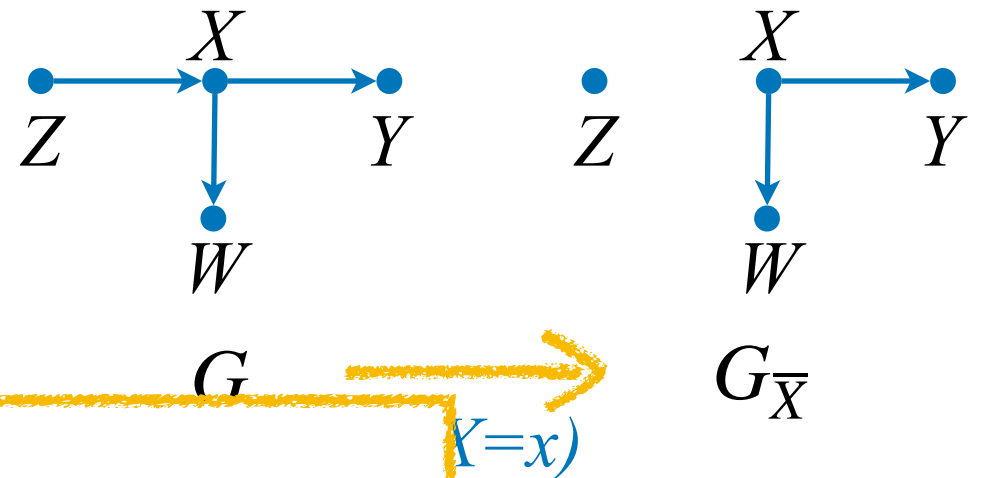
$$= P(z) \neq P(z|w) \quad (Z \not\perp\!\!\!\perp X | W)_{G_{\bar{X}(W)}}$$



in almost any model
compatible with G .

A curiosity about R3

- Verifying whether $P(z|do(x), w) \neq P(z|w)$ holds:

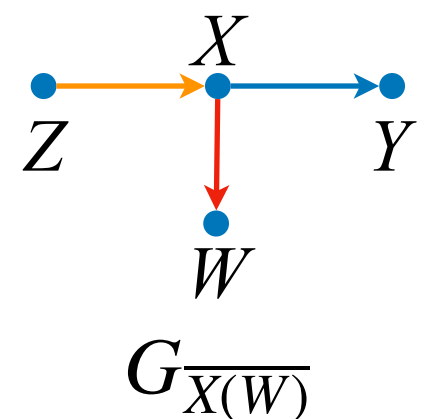


Intuitively:

(i) W is a descendant of X

(ii) in a $do(x)$ -world, conditioning on W does not propagate back to $an(X)$, since incoming arrows into X are removed in the $do(x)$ -world

$\implies$ (iii) when conditioning on W in normal world, conditioning on W **does** propagate to $an(X)$



makes the 3rd rule more subtle!

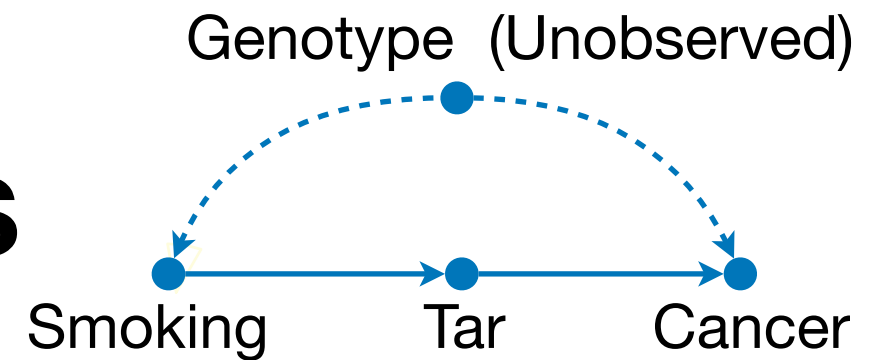
Properties of Do-Calculus

Theorem (soundness & completeness of do-calculus for interventional ID from obs. data).

The quantity $Q = P(y|do(x))$ is identifiable from $P(v)$ and G if and only if there exists a sequence of application of the rules of do-calculus and the probability axioms that reduces Q into a do-free expression.

Syntactic goal: Re-express original Q without $do()$!

Derivation in Do-Calculus



$$P(c | do(s)) = \sum_t P(c | do(s), t) P(t | do(s))$$

$$= \sum_t P(c | do(s), do(t)) P(t | do(s))$$

$$= \sum_t P(c | do(s), do(t)) P(t | s)$$

$$= \sum_t P(c | do(t)) P(t | s)$$

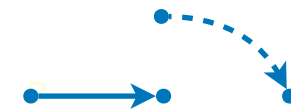
$$= \sum_t \sum_{s'} P(c | do(t), s') P(s' | do(t)) P(t | s)$$

$$= \sum_t \sum_{s'} P(c | t, s') P(s' | do(t)) P(t | s)$$

$$= \sum_t \sum_{s'} P(c | t, s') P(s') P(t | s)$$

Probability Axioms

Rule 2



Rule 2

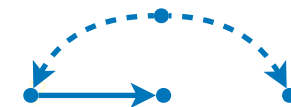


Rule 3



Probability Axioms

Rule 2

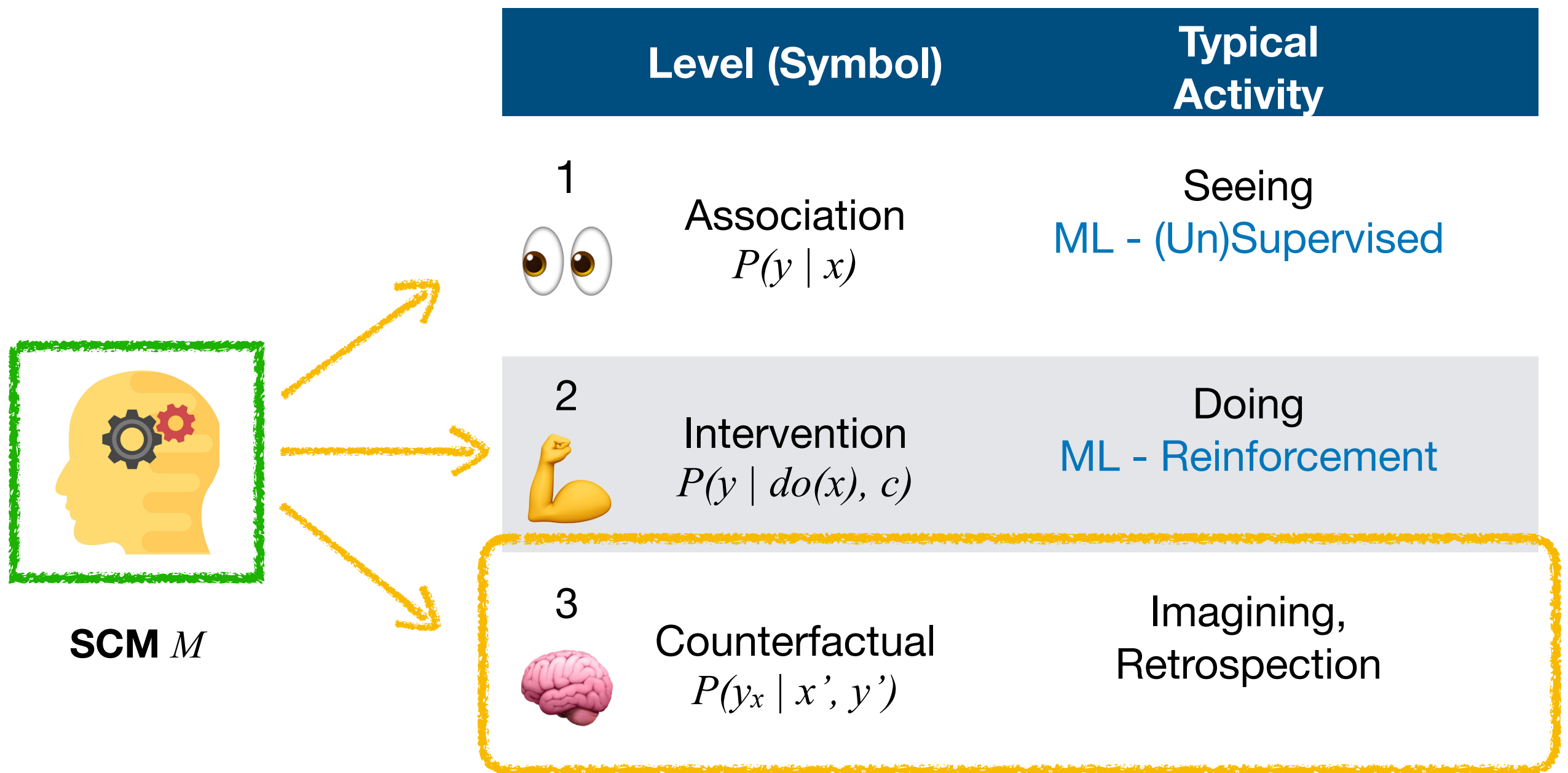


Rule 3



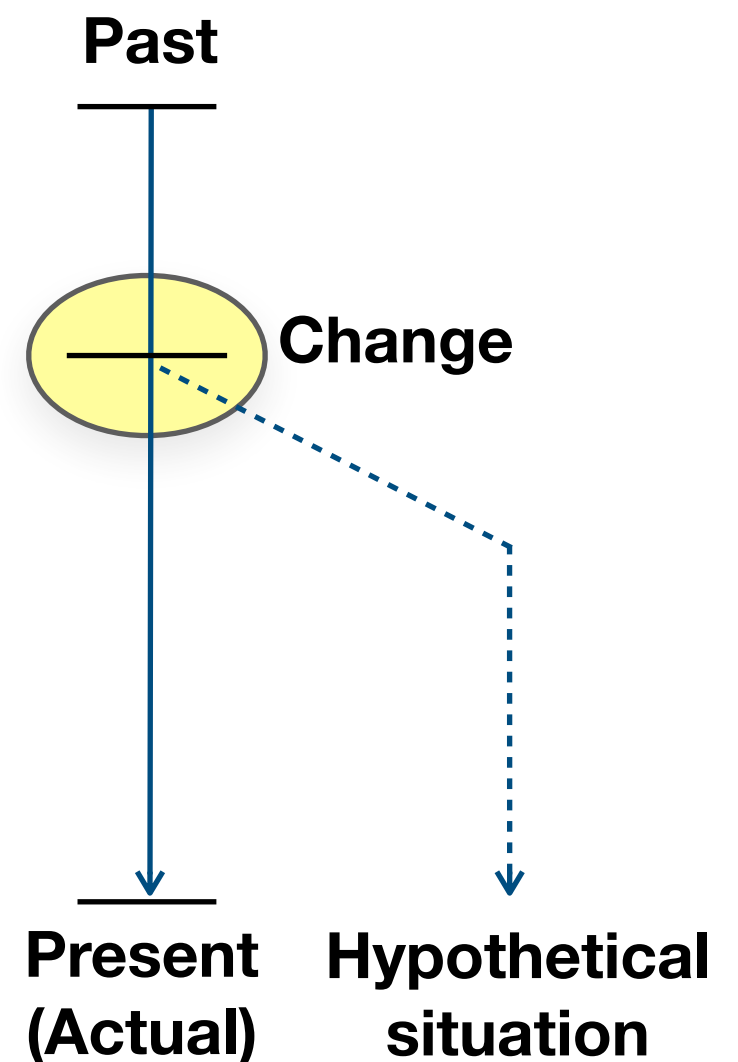
Pearl's Causal Hierarchy— Layer 3: Counterfactuals

SCM → Pearl's Causal Hierarchy

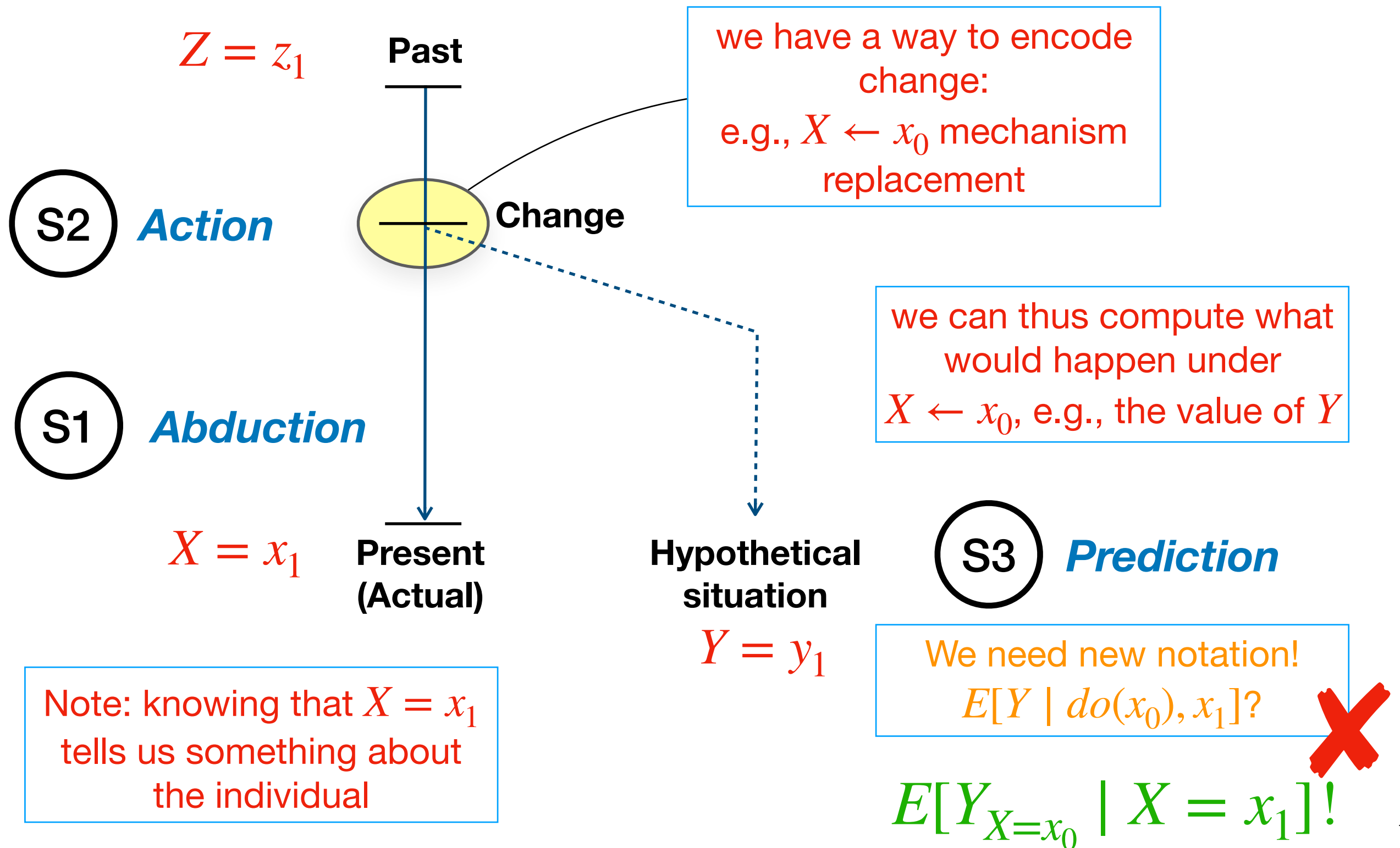


Layer 3 — Semantics

- Layer 3 of the causal hierarchy allows operationalizing the notion of “imagination,” and the closely related activities of retrospection and introspection.
- This entails thinking about alternative ways the world could be, including ways that might conflict with the current state of the world.



Layer 3 – Abduction, Action, Prediction (AAP)



Layer 3 – AAP Example

- Query: $E[Y_{X=x_0} \mid X = x_1]$

S1 *Abduction*

Computing $P(U \mid \text{evidence})$:

$$P(U_z, U_x \mid X = 1) = \frac{P(U_z, U_x, X = 1)}{P(X = 1)}$$

0.2×0.7
 $+$
 0.8×0.1

$$= \frac{P(U_z)P(U_x)P(X = 1 \mid U_z, U_x)}{P(X = 1)}$$

Distribution $P(u \mid x_1)$!

U_z	U_x	$P(U_z, U_x, x_1)$
0	(0, 0)	0
0	(0, 1)	0
0	(1, 0)	$0.8 * 0.1 * 0.4$
0	(1, 1)	$0.8 * 0.1 * 0.6$

U_z	U_x	$P(U_z, U_x, x_1)$
1	(0, 0)	0
1	(0, 1)	$0.2 * 0.9 * 0.7$
1	(1, 0)	0
1	(1, 1)	$0.2 * 0.1 * 0.7$

$\mathcal{M}$

$Z \leftarrow U_z$

$X \leftarrow (U_{x_1} \wedge \neg Z) \vee (U_{x_2} \wedge Z)$

$Y \leftarrow 1(0.8 - 0.3Z + 0.2XZ - 0.1X(1 - Z) < U_y)$

$U_z \sim B(0.2), U_{x_1} \sim B(0.1),$

$U_{x_2} \sim B(0.7), U_Y \sim \mathbf{Unif}[0,1]$

Layer 3 – AAP Example

Ⓢ2 *Action*

Ⓢ3 *Prediction (in M_{x_0})*

$$E[Y_{x_0} \mid X = 1]$$

$$= \sum_u E[Y_{x_0} \mid u] P(u \mid x_1)$$

$$= \frac{32}{220} \times 0.8 \quad + \frac{48}{220} \times 0.8 \quad + \frac{126}{220} \times 0.5 \quad + \frac{14}{220} \times 0.5$$

$U_z = 0, U_x = (1,0)$ $U_z = 0, U_x = (1,1)$ $U_z = 1, U_x = (0,1)$ $U_z = 0, U_x = (1,1)$

$$\approx 0.61$$

$\mathcal{M}_{x_0}$

$$Z \leftarrow U_z$$

$$X \leftarrow x_0$$

$$Y \leftarrow 1(0.8 - 0.3Z + 0.2XZ - 0.1X(1 - Z) < U_y)$$

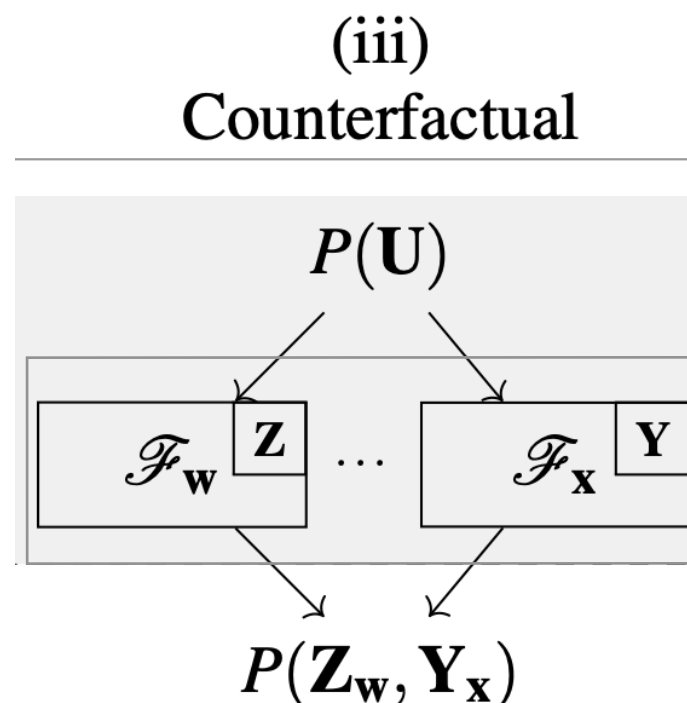
$$U_z \sim B(0.2), U_{x_1} \sim B(0.1),$$

$$U_{x_2} \sim B(0.7), U_Y \sim \mathbf{Unif}[0,1]$$

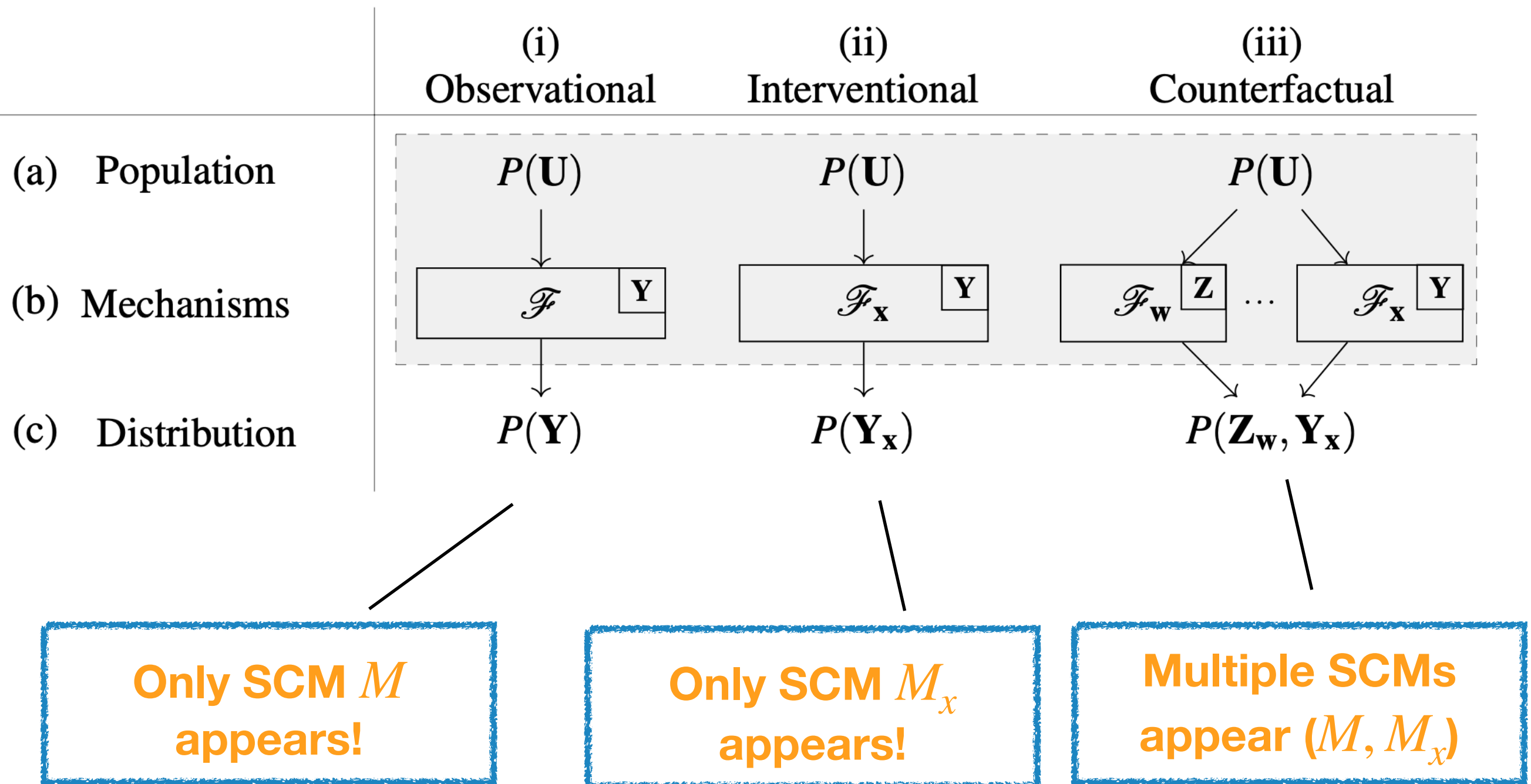
Layer 3 – Valuation

Definition 2.2.6 (Layer 3 Valuation). An SCM $\mathcal{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$ induces a family of joint distributions over counterfactual events $Y_x, \dots, Z_w$, for any $Y, Z, \dots, X, W \subseteq \mathbf{V}$:

$$P^{\mathcal{M}}(\mathbf{y}_x, \dots, \mathbf{z}_w) = \sum_{\mathbf{u}} \mathbf{1} \left(\mathbf{Y}_x(\mathbf{u}) = \mathbf{y}, \dots, \mathbf{Z}_w(\mathbf{u}) = \mathbf{z} \right) P(\mathbf{u}).$$



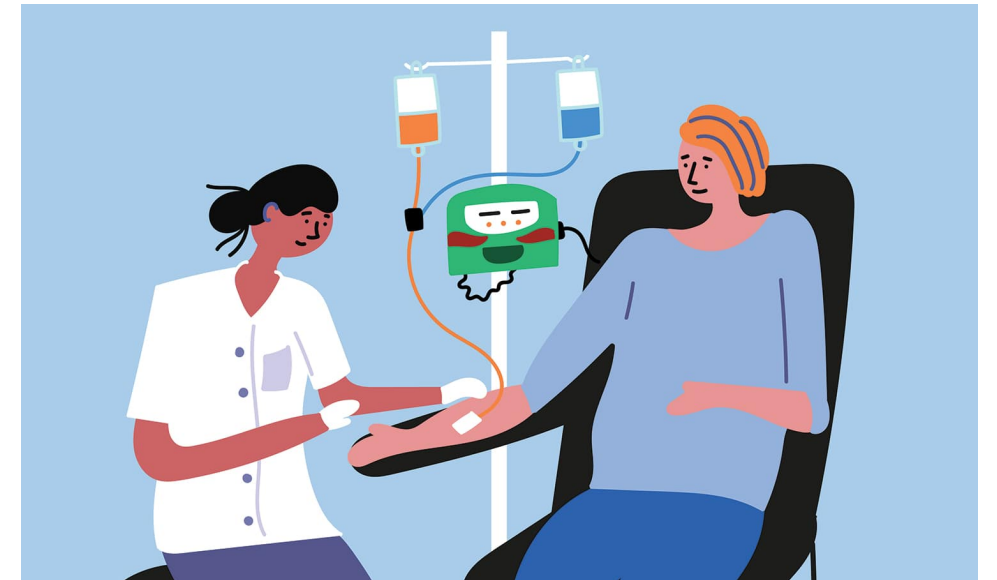
Layer 3 vs. the other layers



II. Why Use Counterfactuals (Motivation)

Example 5.1. Oncologist and Chemotherapy

- An oncologist needs to decide whether to treat her patients, suffering from breast cancer, with chemotherapy,
- For this purpose, she wants to analyze the data collected in her hospital,
- Let X represent the treatment, where $X = 0$ encodes no chemotherapy, and $X = 1$ for chemotherapy.
- Let Y represent the patient's health, where $Y = 0$ means that the patient has died, and $Y = 1$ that the patient is still alive two years after taking the treatment; Z represents how advanced the cancer is ($Z = 0$ for Stage I-II, $Z = 1$ for Stage III-IV).
- Using the data, how should the physician decide to treat her patients?



(1) Observational distribution

- She extracts the preliminary data from the hospital EHR, and gets the observational distribution $P(v)$,

- She then decides to compute the metric $E[Y | X = 1] - E[Y | X = 0]$,

- Based on the table she finds that

$$E[Y | X = 1] = 70 \%$$

$$E[Y | X = 0] = 77.7 \%,$$

- The results imply that treated patients are less likely to survive. At first, they seem puzzling, but...
- Recalling the class in causal inference, she realizes that this quantity is not informative, since data it does not adjusted for confounding bias — more sick patients (Stage III-IV) are more likely to receive chemotherapy in the first place.

P(v)	Z	X	Y
0.144	0	0	0
0.03	1	0	0
0.024	0	1	0
0.576	0	0	1
0.042	1	1	0
0.03	1	0	1
0.056	0	1	1
0.098	1	1	1

(2) Causal Effect Computation

- She then decides to use the adjustment formula based on Z, based on the back-door criterion, $E[y \mid do(x)] = \sum_z E[y \mid x, z]P(z)$

- From BD-adjust, one can compute

$$E[Y \mid do(X = 1)] - E[Y \mid do(X = 0)],$$

- Based on the table she finds that

$$E[Y \mid do(X = 1)] = 70 \%$$

$$E[Y \mid do(X = 0)] = 74 \%,$$

P(v)	Z	X	Y
0.144	0	0	0
0.03	1	0	0
0.024	0	1	0
0.576	0	0	1
0.042	1	1	0
0.03	1	0	1
0.056	0	1	1
0.098	1	1	1

- She finds that even after adjustment, the causal effect of chemotherapy results in worse survival. Does this imply the treatment is futile, and should be abandoned entirely?
- How do make sense of this finding?

(3) Oracle — Effect on Treated

- Let's look at the SCM generating the data, on the left side,
- The physician wants to understand whether the treatment has helped the group of patients who were actually given it,

Would a treated patient ($X=1$) survive without treatment?

$$\begin{aligned} \mathcal{M} \\ Z &\leftarrow \mathbf{Bern}(0.2) \\ X &\leftarrow \mathbf{Bern}(0.1 + 0.6Z) \\ Y &\leftarrow \mathbf{Bern}(0.8 - 0.3Z + \\ &\quad 0.2XZ - \\ &\quad 0.1X(1 - Z)) \end{aligned}$$

- This query has special meaning in the language of layer 3 (L_3) of the PCH, and is written as the counterfactual statement:

$$P(Y_{X=0} = 1 \mid X = 1),$$

This expression is called the *effect of treatment on the treated (ETT)*.

(3) Oracle — Effect on Treated

- What is the probability of survival under a certain treatment, given that the patient's natural treatment regime?
- From the SCM, we can compute:

$$P(Y_{X=1} = 1 \mid X = 1) = 70 \% \quad P(Y_{X=0} = 1 \mid X = 1) = 61 \%$$

- Therefore, among those patients who are treated ($X = 1$ behind the conditioning bar), being treated ($X = 1$) is actually be better since

$$P(Y_{X=1} = 1 \mid X = 1) > P(Y_{X=0} = 1 \mid X = 1),$$

- For those who are not treated, we can show the opposite:

$$P(Y_{X=1} = 1 \mid X = 0) < P(Y_{X=0} = 1 \mid X = 0),$$

- How is this possible?

(3) Oracle – Effect on Treated

- We first look at the mechanism for Y:

$$Y \leftarrow \text{Bern}(0.8 - 0.3Z + 0.2XZ - 0.1X(1 - Z))$$

baseline survival
of 80%

30% survival decrease
for late stage

for late stage,
chemo increases
survival 20%

for early stage,
chemo decreases
survival 10%

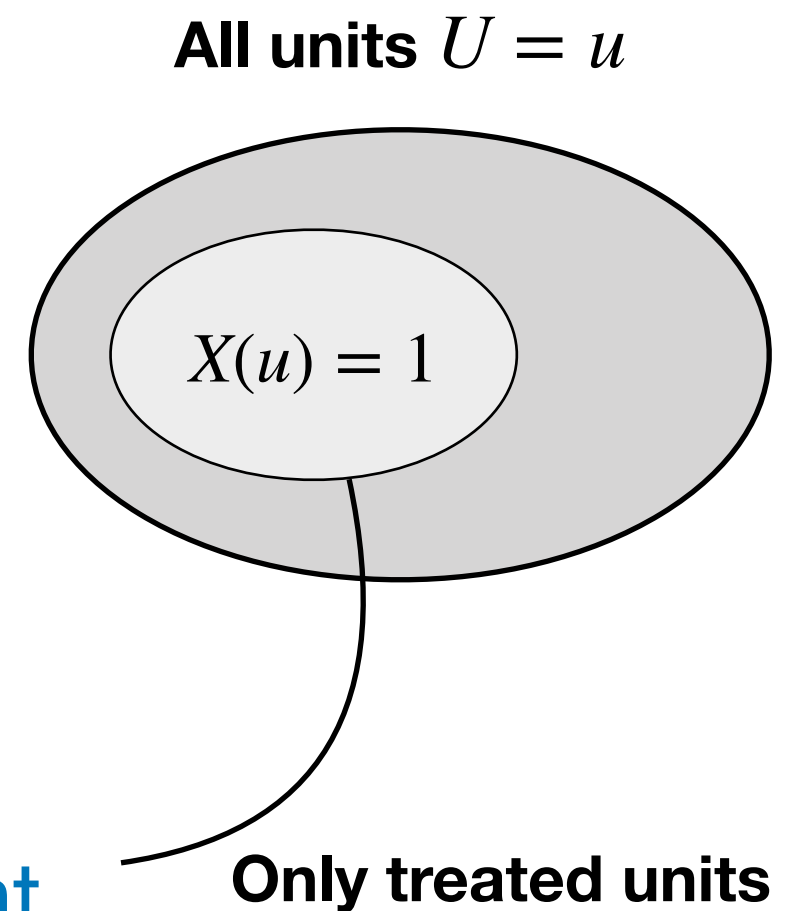
- Therefore, treatment is harmful for early stage, but helpful for late,
- Furthermore, the mechanism for X shows us that

$$X \leftarrow \text{Bern}(0.1 + 0.6Z)$$

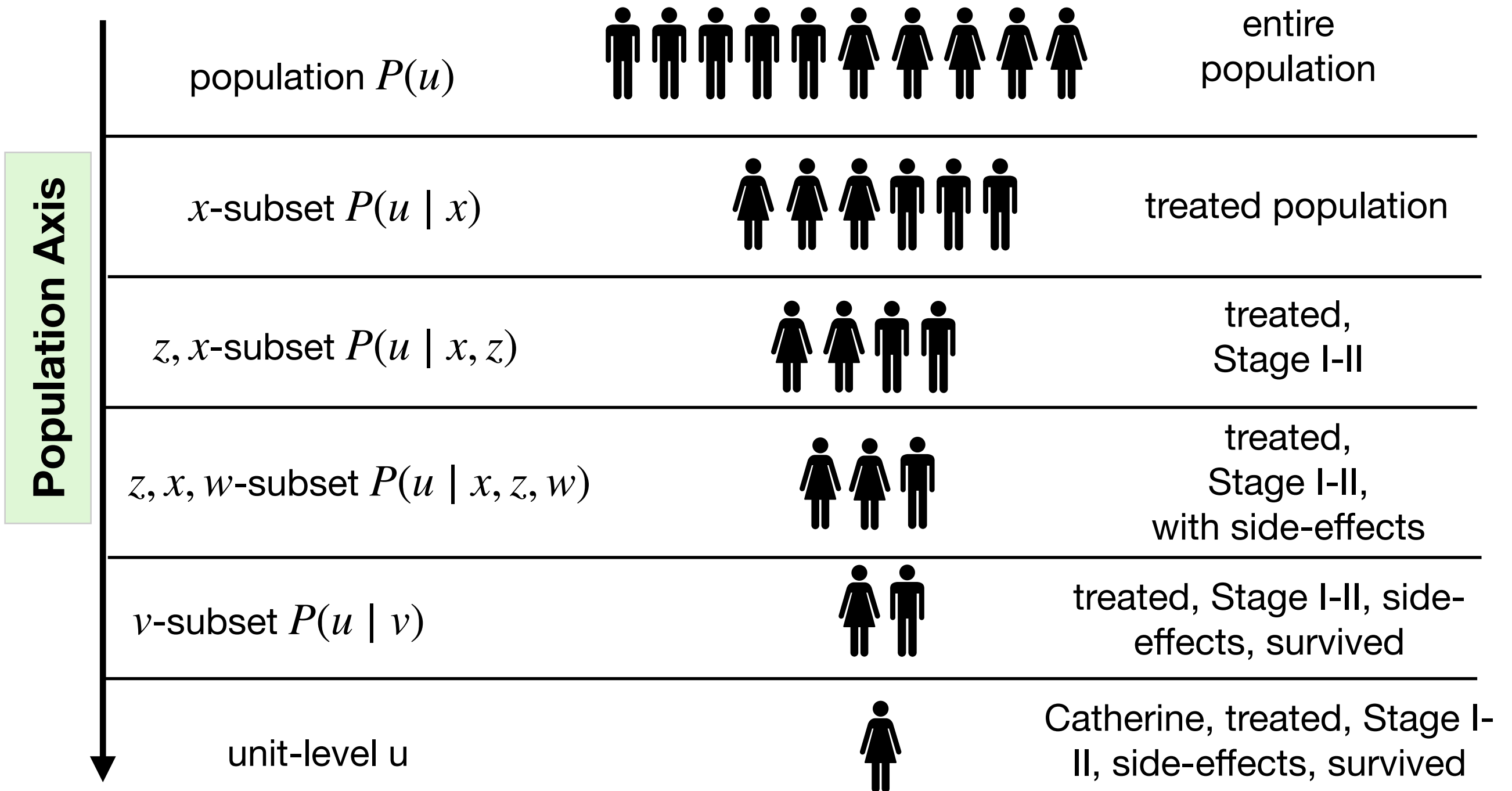
among treated ($X=1$), there are more late-stage patients than usual!

Example – Takeaways

- Causal effects may exhibit *heterogeneity*, meaning that different parts of the population are affected differently,
- The average causal effect $E[Y \mid do(X = 1)] - E[Y \mid do(X = 0)]$ is an average over the entire, possibly heterogeneous, population of individuals,
- To better understand heterogeneity, we may use conditional effects, which sometimes require **Layer 3 of PCH**,
- This is the case for the **Effect of Treatment on the Treated (ETT)**.



(1) Use of Counterfactuals: Granularity



Example. Semaglutide Effects on Diabetes

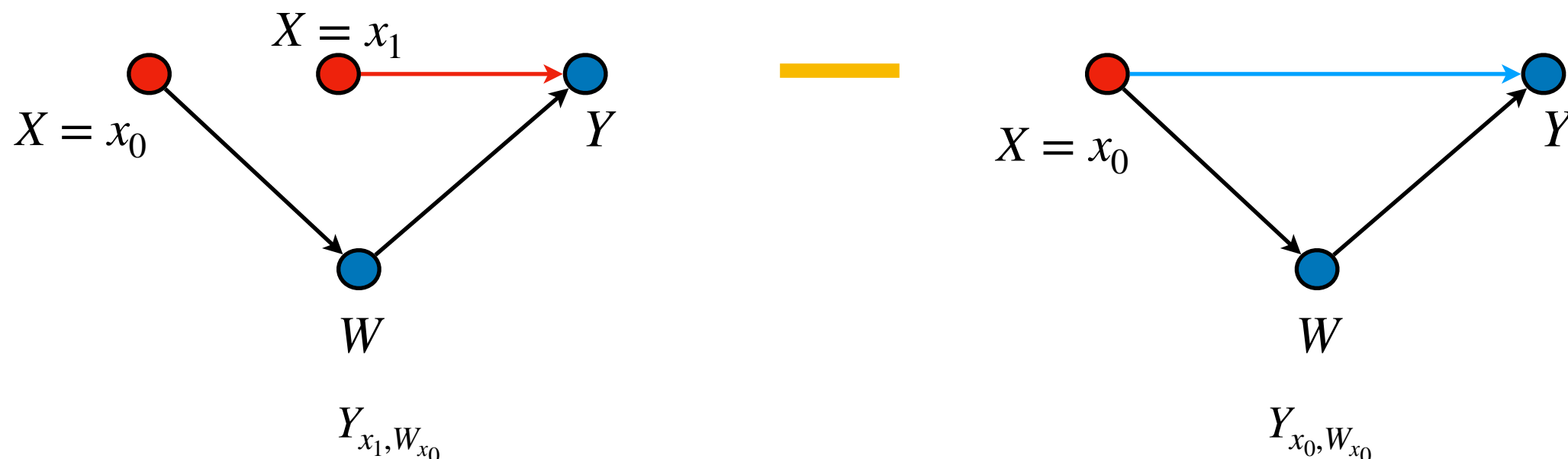
- A diabetologist is treating his patients with semaglutide, to stabilize their HbA1c values,
- However, he realized that there are different causal pathways through which semaglutide may affect HbA1c levels: (1) through weight-loss and associated insulin sensitivity, and (2) direct GLP-1 receptor effects,
- The doctor is interested in how strong each of these pathways is, and he wonders if he can obtain an answer from data:
- $X = 1$ denotes semaglutide treatment, $X = 0$ standard treatment, $Y = 1$ denote normal HbA1c levels ($Y = 0$ otherwise); $W = 1$ for weight loss ($W = 0$ otherwise).



Gedankenexperiment (NDE)

- For investigating this, the doctor performs following thought experiments,
- For an untreated individual ($X = x_0$), how would his HbA1c level (Y) change **had he been** treated ($X = x_1$), while keeping the weight loss unchanged (at the natural level $X = x_0$)?

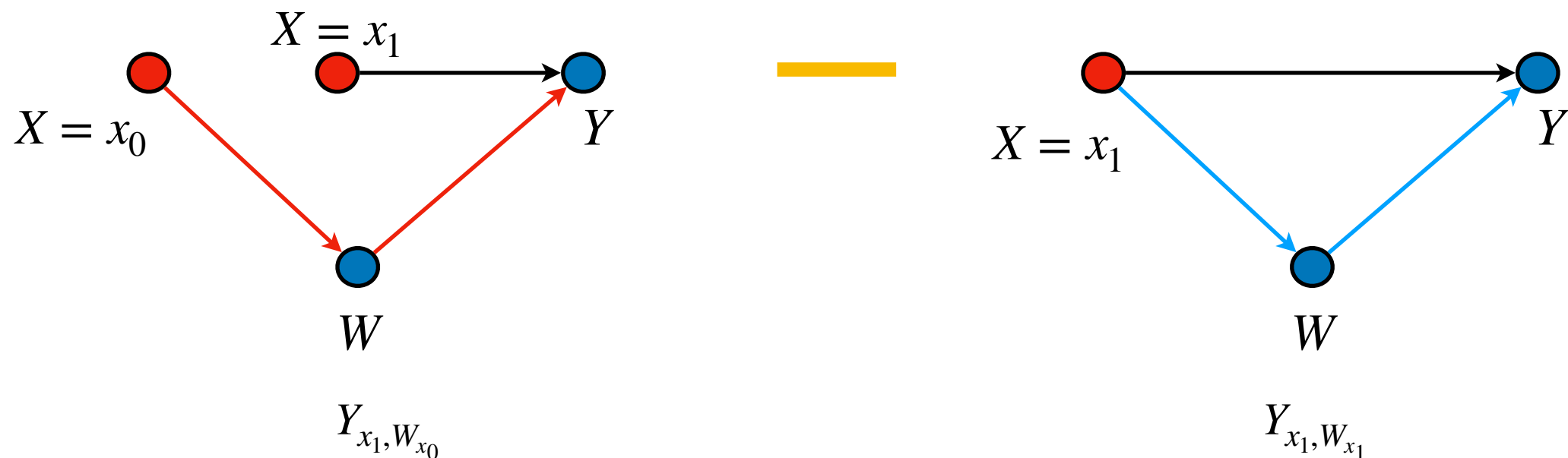
$$\mathbf{NDE}_{x_0, x_1}(y) = P(y_{x_1, W_{x_0}}) - P(y_{x_0, W_{x_0}})$$



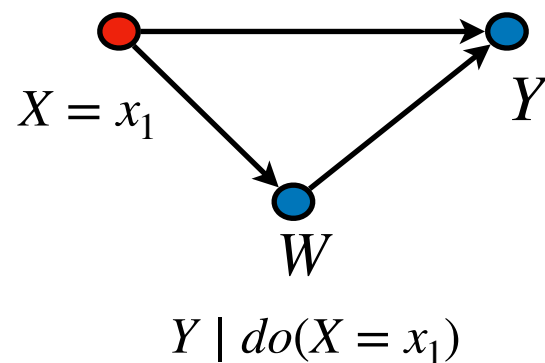
Gedankenexperiment (NIE)

- For a treated individual ($X = x_1$), how would their HbA1c level (Y) change **had they not been** treated ($X = x_0$), while keeping treatment unchanged along the direct causal pathway (at the natural level $X = x_1$)?

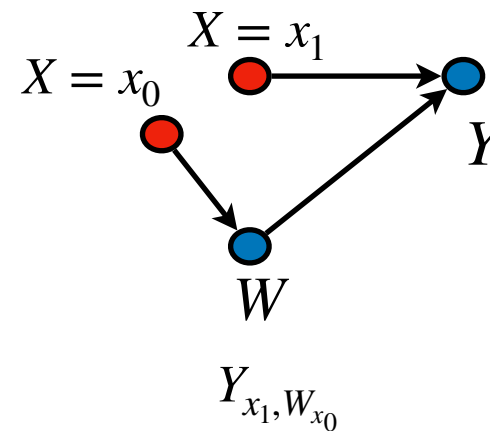
$$\mathbf{NIE}_{x_1, x_0}(y) = P(y_{x_1, W_{x_0}}) - P(y_{x_1, W_{x_1}})$$



ATE Decomposition



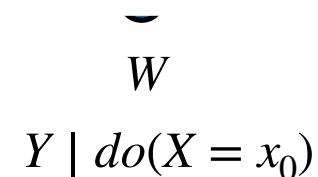
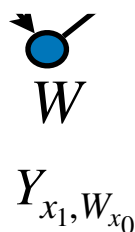
—



— $-NIE_{x_1, x_0}(y)$

Theorem. The average total effect can be decomposed into its direct and indirect parts:

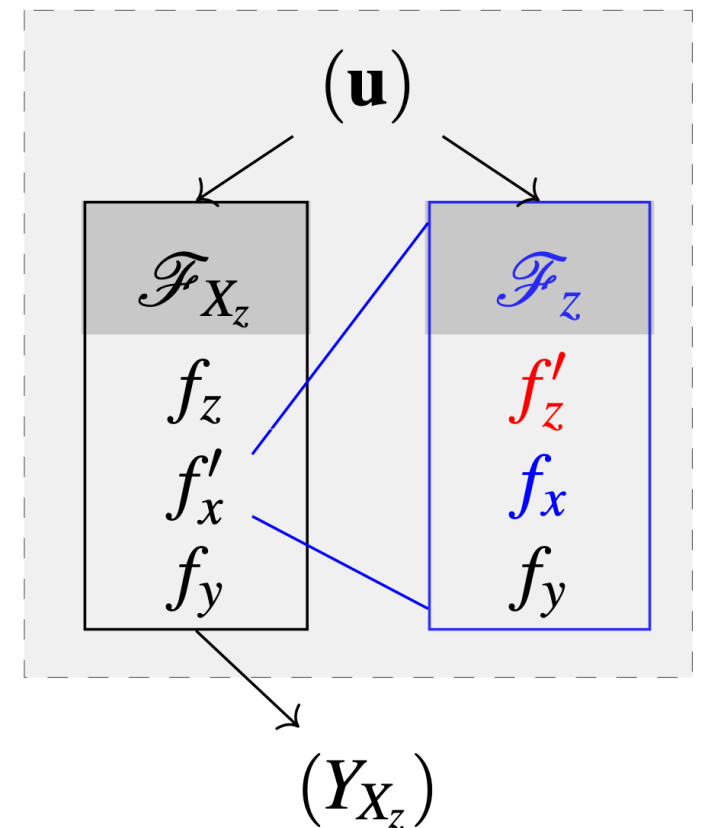
$$ATE_{x_0, x_1}(y) = NDE_{x_0, x_1}(y) - NIE_{x_1, x_0}(y).$$



— $NDE_{x_0, x_1}(y)$

Nested Counterfactuals

- We used the notion of *nested counterfactuals*.
- Y_x refers to Y under intervention $do(X = x)$, that is, when X is fixed to a constant value x .
- Here, we consider an intervention $do(X \leftarrow X_z)$, where X is not fixed to a constant but is supposed to behave as another counterfactual variable X_z .
- For such intervention, we first need to evaluate X_z (from $\mathcal{M}_z$), and then consider a model $\mathcal{M}_{X_z}$ where X is given the value dictated by X_z .

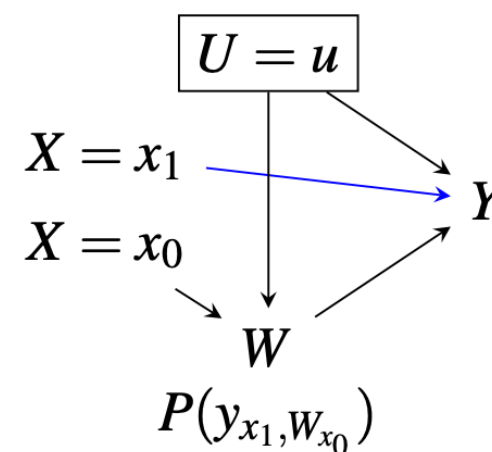
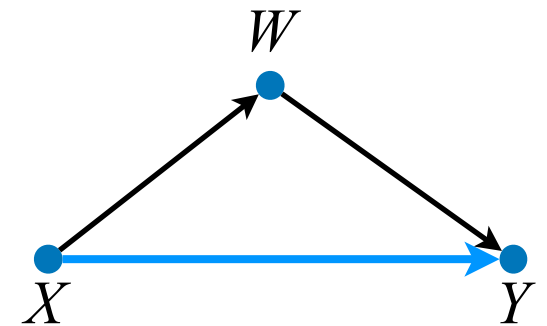


Natural Direct Effects

- The **Natural Direct Effect (NDE)** is defined as

$$\text{NDE}_{x_0, x_1}(y) = P(y_{x_1, W_{x_0}}) - P(y_{x_0, W_{x_0}}) \quad (5.101)$$

- $Y_{x_1, W_{x_0}}$ refers to the outcome Y under $X = x_1$ and $W = W_{x_0}$, the value that W would attain had X been x_0 .
- Y_{x_0} represents a baseline where Y perceives $X = x_0$ in all causal paths. It is also equivalent to $Y_{x_0, W_{x_0}}$, that is, a situation where W also gets $X = x_0$.
- Taking the difference of those two quantities keeps the path $X \rightarrow W \rightarrow Y$ constant while changing the level of X from x_0 to x_1 in the path $X \rightarrow Y$, effectively measuring the direct impact of X on Y .



Summary:

Using of Counterfactuals

Total Effect — TE

$$E[Y \mid do(x_1)] - E[Y \mid do(x_0)]$$

We spent the first 8 lectures building foundations to understand this quantity

Ext. 2: Mechanisms

DE

IE

Ext. 1: Granularity

